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 Studierichting: Informatica
 Oefeningenreeks 3B nr. 1.11

$$\begin{aligned}
 f(x) &= x \operatorname{Bgsin}(x) \\
 f'(x) &= \operatorname{Bgsin}(x) + \frac{x}{\sqrt{1-x^2}} \\
 f''(x) &= \frac{1}{\sqrt{1-x^2}} + \frac{\sqrt{1-x^2} - x \cdot \left(\frac{1}{2\sqrt{1-x^2}} \cdot (-2x) \right)}{(\sqrt{1-x^2})^2} \\
 &= \frac{1}{\sqrt{1-x^2}} + \frac{\sqrt{1-x^2} - \left(\frac{-2x^2}{2\sqrt{1-x^2}} \right)}{1-x^2} \\
 &= \frac{1}{\sqrt{1-x^2}} + \frac{\sqrt{1-x^2} + \frac{x^2}{\sqrt{1-x^2}}}{1-x^2} \\
 &= \frac{1}{\sqrt{1-x^2}} + \frac{(\sqrt{1-x^2})^2 + \frac{x^2\sqrt{1-x^2}}{\sqrt{1-x^2}}}{(1-x^2) \cdot \sqrt{1-x^2}} \\
 &= \frac{1}{\sqrt{1-x^2}} + \frac{1-x^2+x^2}{(1-x^2) \cdot \sqrt{1-x^2}} \\
 &= \frac{1}{\sqrt{1-x^2}} + \frac{1}{(1-x^2) \cdot \sqrt{1-x^2}} \\
 &= \frac{(1-x^2)}{(1-x^2) \cdot \sqrt{1-x^2}} + \frac{1}{(1-x^2) \cdot \sqrt{1-x^2}} \\
 &= \frac{2-x^2}{(1-x^2) \cdot \sqrt{1-x^2}}
 \end{aligned}$$

References